

Homework 2 (20pts)

Why are we doing this? The homework assignments are a low-stakes way to practice the concepts we've talked about in class and apply them to real-world problems. For this particular assignment, I would like you to get more practice working with matrices, and using these matrices to calculate important structural properties of networks.

How will you be assessed? I am looking for *justification* for the mathematical expressions that you derive in Problem 1. For 1(e), you should have corresponding code calculating the various quantities. For problem 2, most of your work will be done in a Jupyter notebook and it should be clear in the notebook that you know what you're calculating through variable names and comments.

What will you submit? Submit a single PDF of the homework assignment and an .IPYNB file with all corresponding code to the Homework 2 Canvas assignment.

Format requirements

- The title of the homework assignment
- Your name
- The collaborators with whom you worked
- Each problem should have a separate heading
- Plots should have descriptive axis labels

The assignment

Extra credit (1 point): Use the LaTeX template (See Canvas) to write up the homework.

Question 1 (11 points): Let $\mathbf{A}$ be the adjacency matrix of an undirected network and $\mathbf{1}$ be the column vector with “1” entries. In terms of these quantities derive expressions for:

- (2 points) The vector $\mathbf{k}$ whose elements are the degrees k_i of the nodes
- (2 points) The number m of edges in the network
- (2 points) The matrix $\mathbf{N}$ whose element N_{ij} is equal to the number of common neighbors of nodes i and j
- (2 points) The total number of triangles in the network, where a triangle means three nodes, each connected by edges to both of the others (Hint: you can use the *trace* of a matrix, which is the sum of its diagonal entries)
- (3 points) From the dolphin social network (See Canvas), create an unweighted adjacency matrix (*Using Python, of course*) by first assigning a “1” if the matrix entry is 5 or larger and zero otherwise and then setting the matrix diagonal equal to 0. Using this post-processed adjacency matrix, the ones vector, and the expressions you derived above to calculate (i) the degree of each node, (ii) the number of edges, (iii) the number of common neighbors for nodes 0 and 3, and (iv) the total number of triangles. Report these as text (No need for tables or figures).

Question 2 (9 points): For this question you will be using the BTSFlights data on Canvas (filename: BTS2025.csv). Below is a code snippet to preprocess the data:

```

df = pl.read_csv("BTS2025.csv")
df = df.filter(pl.col("PASSENGERS") > 1000)

G = nx.DiGraph(df[["ORIGIN", "DEST"]].rows())
lsc = max(nx.strongly_connected_components(G), key=len)
G = G.subgraph(lsc).copy()

```

```

# variables that you may need
nodedict = {node: i for i, node in enumerate(G.nodes)}
reverse_nodedict = {i: node for i, node in enumerate(G.nodes)}
A = nx.to_numpy_array(G, nodelist=nodedict.keys(), dtype=int)

```

This code snippet will load in the data, filter out flights with few passengers (cargo flights), remove the disconnected pieces of the network, create the adjacency matrix, and create dictionaries mapping airport names to indices and vice-versa.

Do the following:

- a) (1 point) Draw the adjacency matrix with a spy plot.
- b) (2 points) Using powers of the adjacency matrix, calculate the diameter of this network. Verify with the `nx.diameter` function.
- c) (2 points) Using powers of the adjacency matrix, find the number of airports that you can reach from Charlottesville (with airport code CHO) by taking (i) one, (ii) two, (iii) three, and (iv) four flights.
- d) (2 points) Calculate the out-degree using the adjacency matrix. What are the top-5 airports in the US by their out-degree and what are their out-degrees?
- e) (2 points) Plot a histogram of the network's out-degree distribution, with enough resolution that you can see interesting behavior. Why do you think the airport network has this kind of degree distribution?

Rubric (Total: 20 points)

Criterion	Points
Problem 1 (11 pts)	
a) Did you explain how you got to the final expression either in words or mathematically? Was the final expression correct?	2
b) Did you explain how you got to the final expression either in words or mathematically? Was the final expression correct?	2
c) Did you explain how you got to the final expression either in words or mathematically? Was the final expression correct?	2
d) Did you explain how you got to the final expression either in words or mathematically? Was the final expression correct?	2
e) Did you correctly create the matrix for the dolphin social network? Did you code up the expressions correctly using Python and Numpy? Were the final answers correct?	3
Problem 2 (9 pts)	
a) Did you correctly draw the spy plot for the adjacency matrix?	1
b) Did you write a function or code that will calculate the diameter? Is the diameter you calculated correct?	2
c) Did you correctly use powers of the adjacency matrix to calculate the number of paths? Did you correctly relate this to the Charlottesville airport?	2
d) Did you correctly calculate the out-degree using the adjacency matrix? Did you correctly relate this back to airports?	2
e) Did you correctly plot a histogram of the out-degrees? Did you relate the shape of this histogram to network concepts we learned in class (i.e., degree distributions)?	2
Total	20